

Wazuh Active Response Incident Report

Findings

Wazuh detected multiple failed SSH authentication attempts against the monitored Linux endpoint. The activity originated from a single source IP address and matched the configured brute-force detection rule.

After the configured detection threshold was reached, Wazuh Active Response automatically executed the configured firewall-based IP blocking action.

The activity was classified as a **True Positive — Simulated SSH Brute-Force Attack**.

Investigation Summary

A simulated SSH brute-force attack was performed against the monitored Linux endpoint in the Wazuh SOC home lab.

Multiple unsuccessful SSH authentication attempts were recorded within a short period. Wazuh analyzed the authentication logs and generated a security alert after the configured detection rule was triggered.

The Wazuh Active Response mechanism then automatically blocked the source IP address using the configured firewall response.

This demonstrated an automated:

Detection → Alert → Response → Containment

workflow.

Who, What, When, Where, Why, How

Who — Who was involved?

- **Source:** Simulated attacker machine
- **Source IP:** <ATTACKER_IP>
- **Target:** <LINUX_HOSTNAME>
- **Target IP:** <TARGET_IP>
- **User targeted:** <USERNAME>

The activity was intentionally generated as part of the security monitoring lab.

What — What happened?

Multiple failed SSH authentication attempts were detected against the Linux system.

Wazuh identified the repeated authentication failures as potential brute-force activity and generated a security alert.

The configured Active Response subsequently blocked the source IP.

When — When did this occur and is it still happening?

- **First observed:** <DATE/TIME>
- **Last observed:** <DATE/TIME>
- **Detection time:** <DATE/TIME>
- **Response time:** <DATE/TIME>
- **Timezone:** IST (UTC+05:30)
- **Current status:** Activity stopped after the source IP was blocked.

Where — Where in the environment did this happen?

- **Target hostname:** <HOSTNAME>
- **Target IP:** <TARGET_IP>
- **Operating System:** Linux
- **Service:** SSH
- **Port:** 22
- **Wazuh Agent:** <AGENT_ID>
- **Source/Attacker IP:** <ATTACKER_IP>

Why — Why did this happen?

The activity was intentionally generated to test the Wazuh SOC home lab's ability to detect and automatically respond to SSH brute-force behavior.

In a real-world incident, the likely objective would be unauthorized access to the SSH service through credential guessing.

How — How did this happen?

The simulated attacker generated multiple unsuccessful SSH authentication attempts against the target Linux server.

The investigation followed this workflow:

SSH Login Attempts → Linux Authentication Logs → Wazuh Agent → Wazuh Manager → Log Decoding → Wazuh Detection Rule → Brute-Force Alert → Active Response → Firewall Rule → Source IP Blocked

MITRE ATT&CK Mapping

- **Tactic:** Credential Access
 - **Technique:** Brute Force
 - **Technique ID:** T1110
 - **Sub-technique:** T1110.001 — Password Guessing (*if applicable to the exact simulation*)
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Indicators of Compromise (IOCs)

Type	Indicator	Description
Source IP	<ATTACKER_IP>	Origin of SSH authentication attempts
Target IP	<TARGET_IP>	Monitored Linux endpoint
Service	SSH	Targeted authentication service
Port	22	SSH service port
Username	<USERNAME>	Account targeted during simulation

Wazuh Evidence

- **Wazuh Agent:** <AGENT_NAME/ID>
- **Rule ID:** <RULE_ID>
- **Alert Level:** <LEVEL>
- **Timestamp:** <TIMESTAMP>
- **Log Source:** /var/log/auth.log (*if applicable*)
- **Event:** Multiple failed SSH authentication attempts
- **Active Response:** Firewall/IP blocking
- **Response Status:** Successfully executed

Evidence Collected

1. Screenshot of failed SSH authentication attempts
 2. Screenshot of the Wazuh alert
 3. Screenshot showing the source IP identified by Wazuh
 4. Screenshot showing Active Response execution
 5. Screenshot showing the firewall rule/IP block
 6. Screenshot of a subsequent connection attempt failing after the IP was blocked
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Response Actions

- [x] SSH brute-force activity detected

- [x] Wazuh alert generated
 - [x] Source IP identified
 - [x] Active Response triggered
 - [x] Source IP automatically blocked
 - [x] Firewall rule verified
 - [x] Post-response connection attempt tested
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Conclusion

Classification: True Positive — Simulated SSH Brute-Force Attack

Wazuh successfully detected repeated failed SSH authentication attempts and automatically initiated the configured Active Response.

The source IP was blocked at the firewall, demonstrating successful automated containment of the simulated attack.

The investigation confirmed that the Wazuh Active Response configuration functioned as intended.

Recommendations

- Implement strong SSH passwords or SSH key-based authentication.
- Disable direct root SSH login where appropriate.
- Restrict SSH access using firewall rules or VPN access.
- Monitor authentication failures continuously.
- Tune the Wazuh brute-force detection threshold to reduce false positives.
- Implement additional detection rules for suspicious authentication patterns.
- Regularly test Active Response to ensure automated containment continues to work.